# PARIMALA DHARSHINI M

**AI Engineer | Python | Machine Learning | Generative AI & LLM Applications**
Chennai, Tamil Nadu | 7810085311 | parimalamurugan15@gmail.com
linkedin.com/in/parimala-dharshini-903b4a271 | github.com/Parimala-15

## PROFESSIONAL SUMMARY

B.Tech Information Technology student (CGPA 9.13/10, graduating 2027) targeting AI engineering roles, with strong Python, machine learning and software engineering fundamentals: Data Structures and Algorithms, Object-Oriented Programming, SQL and databases, debugging and testing. Built machine learning, deep learning and Generative AI projects using Scikit-learn, TensorFlow, PySpark, LangChain and ChromaDB, integrated LLM APIs (Cohere, Gemini) into applications, and developed web applications with Python, Flask, FastAPI, React and MySQL. Completed three internships in Generative AI, machine learning and prompt engineering. Presented a deep learning paper at IEEE QPAIN 2026.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, C, JavaScript, SQL, HTML, CSS

**Machine Learning and Deep Learning:** Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, PySpark, Data Preprocessing, Feature Engineering, Model Training and Evaluation, Cross-Validation, Hyperparameter Tuning, NLP Text Classification, OpenCV, MediaPipe, Matplotlib, Plotly

**AI and Generative AI:** Large Language Models (LLMs), LLM APIs (Cohere, Gemini), Prompt Engineering, LLM Evaluation, RAG, Embeddings, Vector Databases (ChromaDB), Semantic Search, LangChain, LlamaCpp, SentenceTransformers, AI Agents, Tool/Function Calling, MCP

**Software Engineering Fundamentals:** Data Structures and Algorithms (GeeksforGeeks practice), Object-Oriented Programming, Problem Solving, Debugging, Testing, Documentation

**Backend Development, APIs and Databases:** Flask, FastAPI, REST APIs, API Integration, React, Vite, MySQL, SQL (CRUD, JOINs, GROUP BY & Aggregate Functions), Relational Database Design, Normalization

**Developer Tools:** Git, GitHub, Streamlit, Jupyter Notebook, Google Colab, VS Code; AI-assisted development (GitHub Copilot, ChatGPT, Claude, Gemini) with validation of AI-generated output

## PROJECTS

**Cardio-Bot: RAG-Based Healthcare Chatbot** | Python, Streamlit, LangChain, LlamaCpp, ChromaDB, SentenceTransformers, Cohere API

- Built a Retrieval-Augmented Generation (RAG) application in Streamlit: a user question triggers semantic search over a medical knowledge base (SentenceTransformers embeddings, ChromaDB vector database), and the retrieved context is passed to an LLM (Cohere API, LlamaCpp via LangChain) to generate grounded answers.
- Enabled similarity search across 1,000+ medical documents; supports both a hosted LLM API and a local LlamaCpp model.
- Tested with predefined and representative queries, validated responses for relevance and correctness, and debugged retrieval and prompt issues.
- Observed 35% fewer hallucinations and 90% response relevance after refining retrieval and prompts.

**Xeno CRM: Full-Stack CRM with AI Copilot** | React, Vite, FastAPI, Python, MySQL, Gemini API

- Developed a full-stack CRM web application with a React/Vite frontend and a Python FastAPI backend using a MySQL database, organised into Dashboard, Customers, Campaigns, Analytics, AI Copilot and History modules.
- Integrated the Google Gemini API into the AI Copilot module to provide AI-assisted functionality within the CRM workflow (third-party API integration).

**Spectrum Sensing for Cognitive Radio Networks (CLDNN + Attention + Curriculum Learning)** | Python, TensorFlow | IEEE QPAIN 2026

- Designed and trained a deep learning model in TensorFlow combining CNN, Bi-LSTM and Squeeze-and-Excitation attention for signal classification in noisy wireless data.
- Applied a 4-stage cumulative SNR curriculum; trained and evaluated on RadioML 2016.10a across SNR −20 to +18 dB.
- Achieved 73.64% overall accuracy and 0.988 ROC-AUC; presented at IEEE QPAIN 2026, Bangladesh.

**Student Dropout Prediction (OULAD)** | Python, PySpark, Logistic Regression | Big Data Essentials **Nov 2025**

- Merged Open University Learning Analytics Dataset (OULAD) tables into a 213,166-record dataset and labelled dropout as students with a “Withdrawn” final result (30,622 dropouts).
- Trained a Logistic Regression classifier on Spark and evaluated it with cross-validation (78.28% accuracy) and ROC-AUC (0.65).

**Timetable Management System** | Python, Flask, MySQL, SQL, Streamlit

- Designed relational database tables and built a Python (Flask) application backed by MySQL, with SQL-driven CRUD operations for timetable scheduling.
- Used JOINs, GROUP BY and aggregate functions (COUNT, SUM, AVG) to retrieve, organize and analyze timetable data.
- Automated timetable generation, reducing manual scheduling effort by 70%; built an interactive Streamlit dashboard for timetable management.

**Muththamizh: Tamil NLP Chatbot** | Python, NLP, Streamlit

- Developed a Tamil-language chatbot in Streamlit with English–Tamil translation and Tamil literature queries; added text-to-speech, text summarization and word search across PDF/DOCX files.

**Eye Tracker & Blink Detection System** | Python, OpenCV, MediaPipe

- Built a real-time webcam eye-tracking and blink-detection system using OpenCV and MediaPipe; mapped blinks to mouse clicks for hands-free computer interaction.

## EXPERIENCE

**Generative AI Intern** | Infosys Springboard (Remote) **Dec 2025 – Feb 2026**

- Developed an LLM-based personalized cover letter generator in which structured user inputs drive engineered prompts to produce tailored letters.
- Improved relevance and response quality through prompt optimization and comparative LLM evaluation.
- Designed reusable workflows for automated AI-driven content generation.

**Machine Learning Intern** | Innomatics Research Labs (Remote) **Nov 2025 – Mar 2026**

- Built end-to-end ML pipelines in Python covering data preprocessing, feature engineering, model training and evaluation.
- Tuned models with hyperparameter search and cross-validation to improve performance.
- Implemented NLP text-classification workflows in Scikit-learn: text preprocessing, vectorization and classification.

**AI-Powered Data Insights & Prompt Engineering Intern** | Excelerate (Remote) **Apr 2025 – May 2025**

- Built a supervised ML model in Python (Scikit-learn, PySpark) to predict student dropout risk: 88% accuracy and 20% F1-score improvement through feature engineering and hyperparameter tuning.
- Compared Claude and Cohere through prompt engineering experiments, scoring responses on accuracy, usability, scalability and cost, and refined prompts based on the results.
- Built dashboards and data visualizations and documented findings to refine the approach.
- Led project work with engineering, data, sales and marketing teams on real customer use cases.

## EDUCATION

**B.Tech, Information Technology** | SRM Institute of Science and Technology, Chennai **2023 – 2027**

CGPA: 9.13/10 | No pending backlogs | Coursework: Database Management Systems, Big Data Essentials

## CERTIFICATIONS

- Generative AI (GUVI)
- Machine Learning Specialization (Coursera)
- Database Management Systems (NPTEL)
- TCS iON Career Edge – Young Professional

## ACHIEVEMENTS & LEADERSHIP

- Presented a deep learning research paper at the IEEE 2nd International Conference on Quantum Photonics, AI and Networking (QPAIN 2026), Bangladesh.
- Participated in ImpactX 3.0, a 24-hour SDG hackathon (St. Joseph's Institute of Technology); presented the spectrum sensing project as part of a team.
- Practised Data Structures and Algorithms and problem solving on GeeksforGeeks.
- Team leader for academic projects; Marketing Team Member, Project Club.

## LANGUAGES

English – Professional Working Proficiency | Tamil – Native Proficiency | Hindi – Basic/Academic Proficiency